

DAX Math and Trigonometry Functions

A concise, print-ready reference for Power BI analysts and BI developers.

- ❑ This deck documents DAX math and trigonometry functions with definitions, syntax, parameters, return values, examples, and visuals.

Introduction 4 What are Math & Trig Functions in DAX?

Math and trigonometry functions in DAX provide deterministic numeric transformations used in measures, calculated columns, and table expressions. They support arithmetic, rounding, powers, logarithms, roots, and trigonometric operations required for analytics, forecasting, geospatial calculations, and financial measures.

- Purpose: Normalize, transform, and compute numeric values within data models.
- Contexts: Measures, calculated columns, calculated tables, and what-if parameters.
- Precision: Many functions operate on floating-point values & watch for precision artifacts.

This reference organizes functions alphabetically and by category (rounding, exponentials/logs, powers/roots, trigonometry), includes sample DAX, return types, and practical scenarios to integrate into Power BI reports.



Overview Table for Functions at a Glance

Quick index for reference. See detailed cards for full syntax and examples.

Function Name	Purpose	Return Type	Use Case
ABS()	Absolute value of number	Decimal	Normalize negative amounts
ROUND(), ROUNDUP(), ROUNDDOWN()	Round numbers to specified decimals	Decimal	Financial rounding
CEILING(), FLOOR()	Round to nearest multiple up or down	Decimal	Bucket pricing/taxes
INT()	Return integer portion	Integer	Truncate for bins
MOD()	Remainder after division	Integer/Decimal	Periodic patterns, even/odd tests
POWER(), SQRT(), PI()	Powers, square root, Å constant	Decimal	Geometric calculations
EXP(), LN(), LOG(), LOG10()	Exponential and logarithms	Decimal	Growth rates, normalization
SIN(), COS(), TAN(), ASIN(), ACOS(), ATAN(), ATAN2(), RADIANS(), DEGREES()	Trigonometric conversions and ratios	Decimal	Geospatial and angle math



Tip: Use the function index when authoring measures to pick the right return type and avoid implicit conversions.

Core Rounding & Integer Functions



ROUND(number, num_digits)

Definition: Rounds to nearest specified decimal place. num_digits can be negative to round to positions left of decimal.

Return: Decimal.

Example: ROUND([Amount], 2) ³ 1234.57

Use Case: Standard financial display of currency values.



ROUNDUP(number, num_digits)

Rounds away from zero. Useful when you must overestimate or allocate extra.

Example: ROUNDUP(12.301, 2) ³ 12.31



ROUNDDOWN(number, num_digits)

Always rounds toward zero. Use in conservative estimates or truncation workflows.

Example: ROUNDDOWN(12.399, 2) ³ 12.39



FLOOR(number, significance)

Rounds number down to nearest multiple of significance. Example: FLOOR(25, 10) ³ 20.



CEILING(number, significance)

Rounds up to nearest multiple. Example: CEILING(25, 10) ³ 30. Useful for pricing tiers.

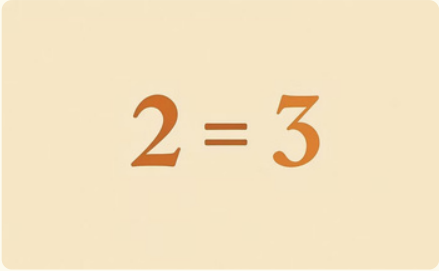


INT(number)

Returns integer portion by rounding down toward negative infinity. INT(-1.2) ³ -2. Use when you require consistent integer binning.

Visual example: Rounding differences often change aggregates 4 test measures over your dataset to confirm expected totals.

Powers, Roots, and Constants

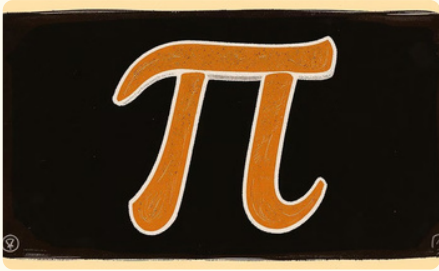

$$2 = 3$$

POWER(number, power) Sqrt(number)

Computes $\text{number}^{\text{power}}$.
Return: Decimal. Example:
POWER([Value], 3) for cubic
transforms used in
compounded growth
scenarios.


$$\sqrt{R}$$

Returns squareroot. Return:
Decimal. Example:
SQRT([Area]) used in
geometric measures and
standard deviation rebuilds.


$$\pi$$

PI()

Returns Å constant (3.14159...).
Use in circle circumference
and geospatial distance
approximations.

Example measure combining them: Circle Area = PI() * POWER([Radius], 2). Test with sample rows to validate units and types.

Exponential & Logarithmic Functions

EXP(number)

Returns e raised to the given power. Useful in continuous growth and decay models.
Example: $\text{EXP}(\text{Rate})$ for continuous compounding.

LN(number)

Natural logarithm (base e). Inverse of EXP.
Example: $\text{LN}(\text{Value})$ to linearize exponential trends.

LOG(number, base) and LOG10(number)

LOG allows arbitrary base. LOG10 specifically returns base-10 log. Use to compress skewed distributions and prepare data for visual scaling.



Example: Convert exponential prediction to rate: $\text{Rate} = \text{LN}(\text{FutureValue} / \text{PresentValue}) / \text{Time}$. Implement as a measure using LN and DIVIDE to avoid division by zero.

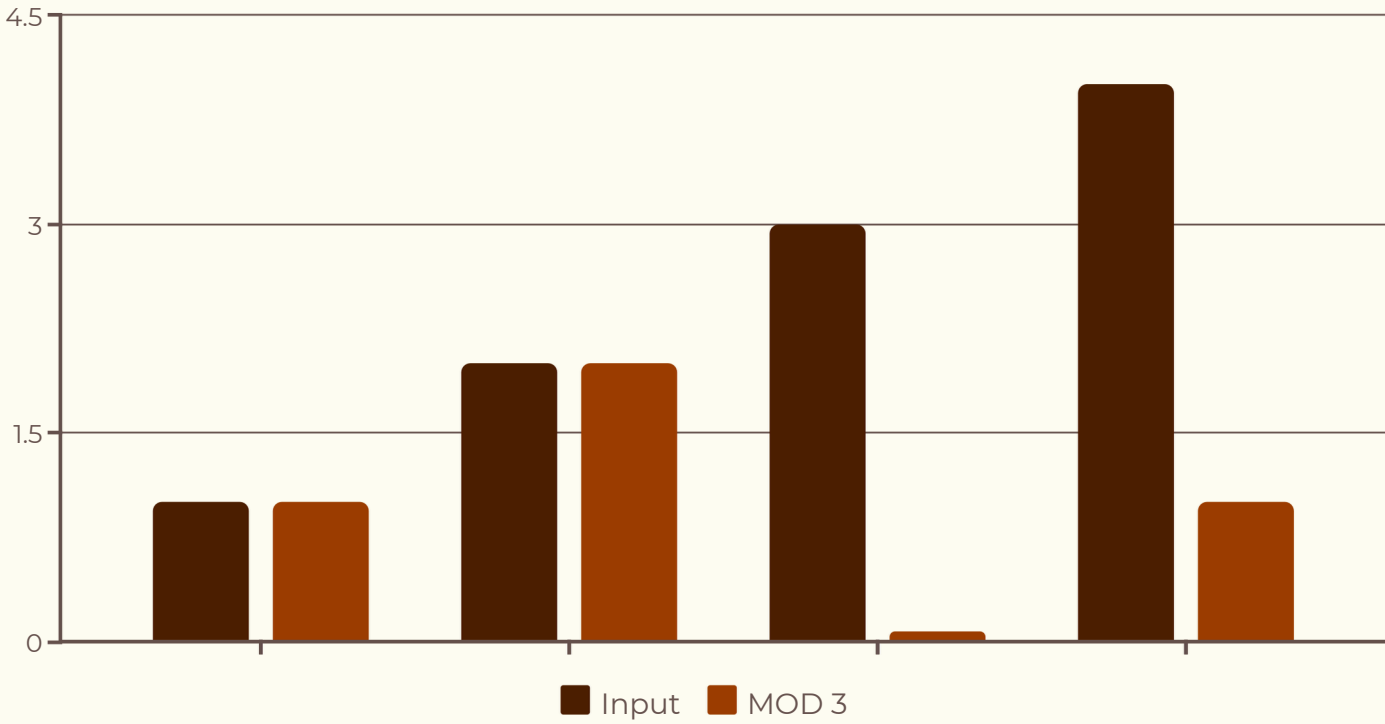


Best practice: wrap divisive formulas with $\text{DIVIDE}(\text{numerator}, \text{denominator}, \text{alternateResult})$ to avoid errors from zero denominators.

Modulo and Remainder

`MOD(number, divisor)` returns the remainder after dividing `number` by `divisor`. Return type: numeric. This function is fundamental for periodic logic (e.g., week-of-month calc, alternating patterns) and parity checks.

Example: `IsEven = IF(MOD([RowNumber], 2) = 0, "Even", "Odd")`



Use Case: Align events into repeating cycles (e.g., assign shift groups by day index using MOD).

Trigonometric Functions & Angle Conversions

SIN(number)



Returnssineofangle in radians. Use RADIANS(degrees) to convert degrees when necessary. Example: SIN(RADIANS([Bearing])) in geospatial formulas.



COS(number)

Returns cosine;used together with SIN for coordinate transforms (lat/long math).



TAN(number)

Returnstangent.Beware: TAN(PI()/2) is undefined 4 guard against angle singularities.



ASIN(), ACOS(), ATAN()

Inverse trig functions returnangle in radians given a ratio. Use DEGREES() to convert to degrees for display.



ATAN2(y, x)

Computesangle from x,y coordinates and resolves quadrant correctly. Essential for bearing calculations: ATAN2(deltaY, deltaX).



RADIANS(), DEGREES()

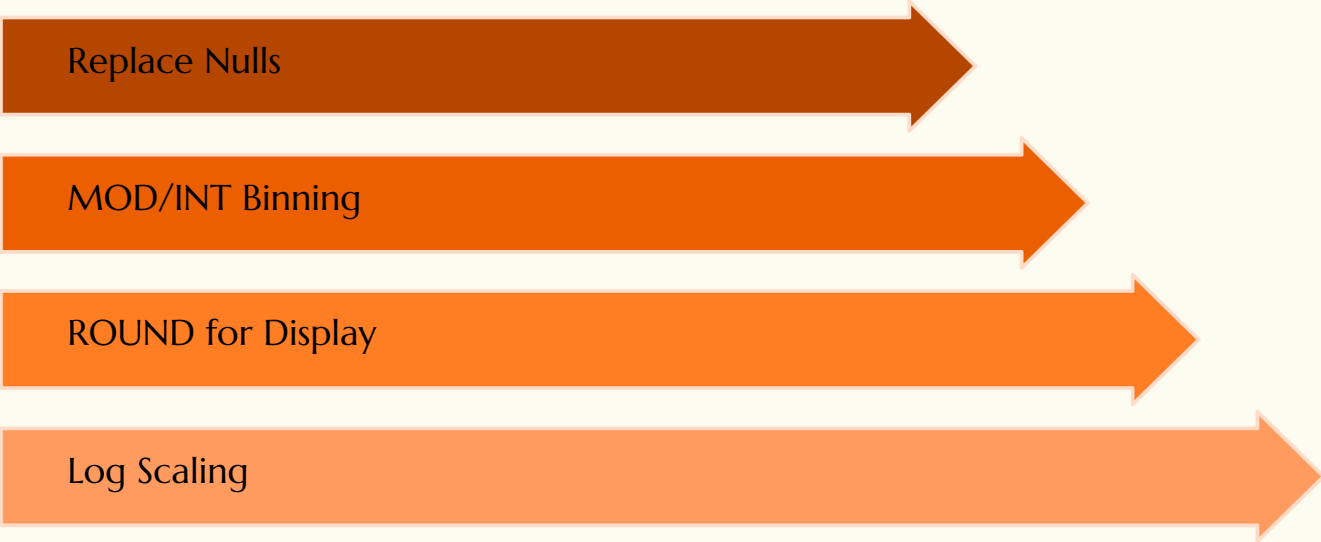
Convertbetween degreesand radians:
RADIANS(30) $\approx$ 0.5236.
DEGREES(PI()/6) $\approx$ 30.

Comparisons, Examples & Best Practices

Compare ROUND, ROUNDUP, and ROUNDDOWN to choose the right behavior for your measure. Use INT when you need truncation semantics that differ for negative values.

Function	Input	Result
ROUND(12.345, 2)	12.345	12.35
ROUNDUP(12.341, 2)	12.341	12.35
ROUNDDOWN(12.349, 2)	12.349	12.34
INT(-1.2)	-1.2	-2

- Precision: Use ROUND to present numbers; use explicit data type casts if required by model consumers.
- Stability: Use DIVIDE() and error-handling patterns when combining logs or ratios.
- Testing: Add unit-test sample tables (small calculated table) to validate measure behavior across edge cases (zero, negative, extreme magnitudes).



Summary, References & Author

Key Takeaways

- Choose the function that matches numeric intent: rounding vs truncation vs bucketing.
- Wrap risky operations (division, logs, trig singularities) with guards (DIVIDE, IF, error handling).
- Use ATAN2 for coordinate-based angles and always manage unit conversions with RADIANS/DEGREES.
- Validate results on sample datasets to catch floating-point edge cases early.

For production measures, document assumptions (units, decimal precision) in measure descriptions and maintain a test dataset to ensure reproducibility.



References:

1. Microsoft Learn 4 DAX Math & Trig Functions (official docs).
2. Power BI Documentation 4 Best practices for modeling and measures.

Created by Your Name | Based on Microsoft Learn (2025)

